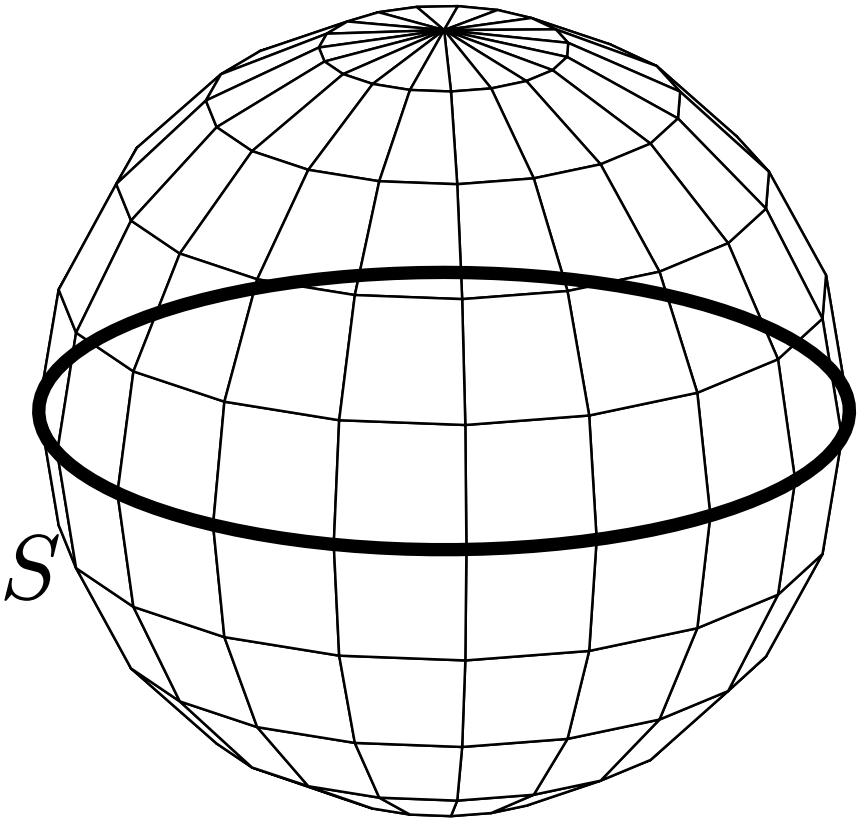$$S^2 \subset S^3 \subset \dots$$